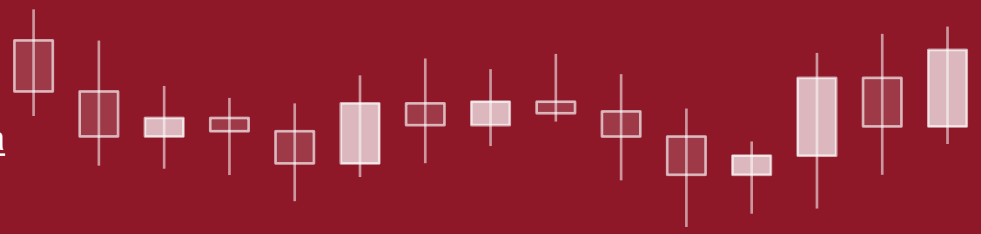
AI, MACHINE LEARNING AND BIG DATA FOR FINANCE AND MANAGEMENT · FINAL REPORT

Can Machine Learning Trade the Next Nasdaq Candle?

Next-candle direction models and long/short backtests on E-mini Nasdaq-100 futures (NQ), after commissions

Alessandro Pacifici
Waseda University · November 2026
Option 1: data analysis · Code and results: github.com/Paciooo/Project_ai_ML_bigdata



Executive Summary

Models predict slightly better than chance, but no strategy survives costs and robustness tests

6.2M

one-minute NQ bars (2006–2025), tested on 1, 5, 15 and 60-minute candles

+0.62 pp

best accuracy gain over “always up” (MLP 128-64-32, 5 min, 95% CI 0.32 to 0.92)

349

model × direction × filter strategies compared on validation data

0

strategies passing all tests: profitable out of sample, after costs, and robust

Problem

Can a model predict whether the next Nasdaq futures candle closes above its open, and trade that signal profitably after $5 round-trip commissions?

Approach

Original vs optimised logistic regression, XGBoost, random forest and neural networks (up to 4 hidden layers), new causal features, strict chronological splits, long/short, long-only and short-only backtests.

Verdict

Optimisation adds at most a few tenths of a point of accuracy. Unfiltered strategies lose heavily after costs; filtered ones hover around zero. No deployable strategy was found.

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Task, Data and Models

Predict the body of the next candle at the close of the current one, then trade exactly that candle

Prediction task (no look-ahead)



- **Target:** bullish if close(t+1) > open(t+1), bearish if lower; doji = miss
- **Data:** NQ futures (CME E-mini Nasdaq-100, $20 per point), TradeStation 1-min bars 2006–2025
- **Splits:** train 2006–Aug 2017 · validation → Jul 2021 · test Jul 2021–May 2025, purged at boundaries

Model families: original and optimised versions

1. **Logistic regression**
   weighted sum of features → probability of an up candle; simple, hard to overfit
2. **Random forest**
   averages hundreds of decision trees built on random data subsets
3. **XGBoost**
   trees added one by one, each correcting the previous errors; early-stopped on validation
4. **Neural network (MLP)**
   stacked layers of neurons learning non-linear patterns; 2 to 4 hidden layers tested

Features: 8 original price features (lagged returns, momentum, volatility) + 11 new ones (candle body and wicks, range, volume, realised volatility, RSI, moving-average gaps, time of day).

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Original vs Optimised Models

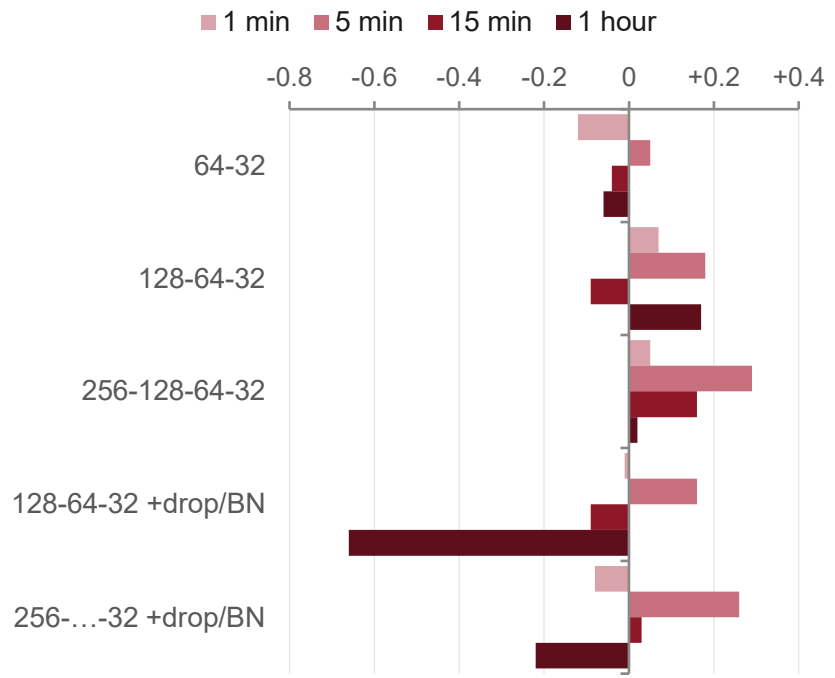
Optimisation changes out-of-sample accuracy by less than one point; deeper networks give no consistent gain

Test accuracy, all candles (original → optimised, change in pp)

Model	1 min	5 min	15 min	1 hour
Logistic regression	48.63% → 48.67% +0.04 pp	50.42% → 50.43% +0.01 pp	50.73% → 50.77% +0.05 pp	51.09% → 50.50% −0.59 pp
Random forest	48.68% → 48.69% +0.01 pp	50.49% → 50.47% −0.02 pp	50.88% → 50.79% −0.09 pp	51.11% → 51.69% +0.59 pp
XGBoost	48.63% → 48.63% −0.00 pp	50.50% → 50.46% −0.04 pp	50.67% → 50.96% +0.29 pp	50.77% → 51.76% +0.98 pp
Neural network (MLP)	48.66% → 48.71% +0.05 pp	**50.33% → 50.52% +0.18 pp**	50.94% → 50.85% −0.09 pp	51.57% → 51.74% +0.17 pp

Highlighted: best predictive model, MLP 128-64-32 on 5 min candles: 51.45% correct on non-doji candles, +0.62 pp vs always-up (95% CI 0.32 to 0.92). Dojis count as misses, so all-candle accuracy at 1 min is below 50%.

Neural networks: test accuracy vs original 64-32-16, pp



Early stopping halted every network after 5–17 epochs: validation loss stops improving almost immediately, a sign of a very weak signal. More neurons or layers gave no consistent gain; dropout/batch norm did not help.

Accuracy beats "always up" only at 1 and 5 minutes; from 15 minutes on, no model is reliably better

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From Signals to Trades

Every unfiltered strategy loses money once each round trip pays $5; costs are larger than the gross edge

Trading rules (one NQ contract)

1 **Signal**
model probability that candle t+1 is bullish

2 **Direction**
Long & Short · Long only · Short only

3 **Filter (optional)**
trade only if |P(up) − 0.5| ≥ threshold chosen on validation

4 **Execution**
enter at open(t+1), exit at close(t+1); P&L = ±(close − open) × $20

5 **Costs**
$5 per round trip ($2.50 open + $2.50 close), zero slippage, 100% fills (idealised)

Validation-chosen model per candle length, Long & Short, test period

Candle	Model	Trades	Gross	Costs	Net	Filtered trades	Filtered net
1 min	MLP 256-128-64-32	1,370,692	−$235.0k	−$6.85M	**−$7.09M**	2,163	**+$30.7k**
5 min	MLP 128-64-32, lr 3e-4	273,340	+$114.9k	−$1.37M	**−$1.25M**	9,717	**−$57.4k**
15 min	MLP 128-64-32, lr 3e-4	90,189	−$47.9k	−$450.9k	**−$498.8k**	1,082	**−$2.4k**
1 hour	XGBoost d5, early stop	21,507	+$32.6k	−$107.5k	**−$74.9k**	17,419	**−$7.6k**

Filter thresholds (validation): 1 min 0.1 · 5 min 0.05 · 15 min 0.05 · 1 hour 0.01. The filter cuts the number of trades by 19% to 99.8% and costs with it; what remains is close to zero and unreliable.

Example, 5-minute reversal rule: +$391.3k gross over 273,340 trades becomes −$975.4k after commissions: $1.43 gross edge per trade vs $5.00 cost

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Backtest by Model: 1 hour Candles

After costs only one of 15 model × direction combinations is positive, and only marginally

Test period, no filter, one contract, $5 per round trip (1 hour candles chosen on validation)

Model	Accuracy	Net: Long & Short	Net: Long only	Net: Short only	Win rate	Trades	Profit factor	Max drawdown	Net / trade
Logistic C=1 + features	50.50%	−$128.9k	−$89.2k	−$39.8k	50.1%	21,507	0.98	−$160.8k	−$6
XGBoost d5, early stop	51.76%	−$74.9k	−$69.1k	−$5.8k	51.3%	21,507	0.99	−$150.9k	−$3
Random forest d10	51.69%	−$5.2k	−$31.0k	+$25.7k	51.3%	21,507	1.00	−$113.2k	−$0
MLP 128-64-32	51.74%	−$130.0k	−$92.5k	−$37.6k	51.3%	21,507	0.98	−$180.9k	−$6
Reversal rule	50.52%	−$153.6k	−$68.2k	−$85.4k	50.1%	21,507	0.97	−$257.8k	−$7
Always long	51.53%	—	−$93.6k	—	51.1%	21,507	0.98	−$169.7k	−$4

Win rate, trades, profit factor, drawdown and net per trade refer to the Long & Short version (Long only for “always long”). Highlighted row: the model of the pre-specified best strategy.

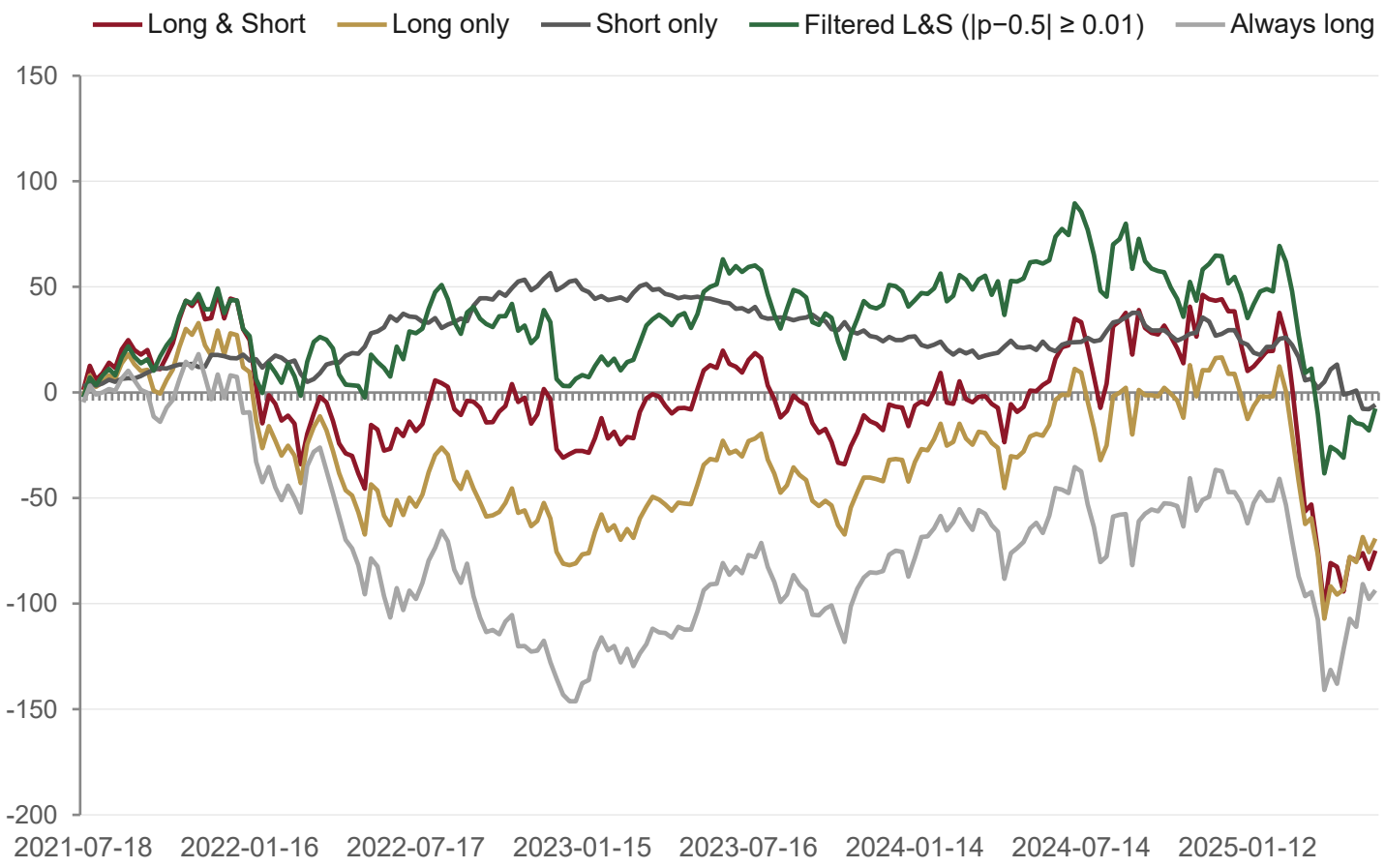
Higher accuracy does not mean profit: the most accurate models are still down after costs, because wins and losses are of similar size and every trade pays $5

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Equity Curves: Long vs Short

Long trades dominate the losses; short-only loses least but is still negative after costs

Cumulative net P&L, XGBoost d5, early stop on 1 hour candles, test period (thousand USD)



Long vs short contribution (no filter)

	Long only	Short only
Trades	18,489	3,018
Net P&L	−$69.1k	−$5.8k
Win rate	51.5%	50.0%
Profit factor	0.99	0.99
Max drawdown	−$140.0k	−$65.8k

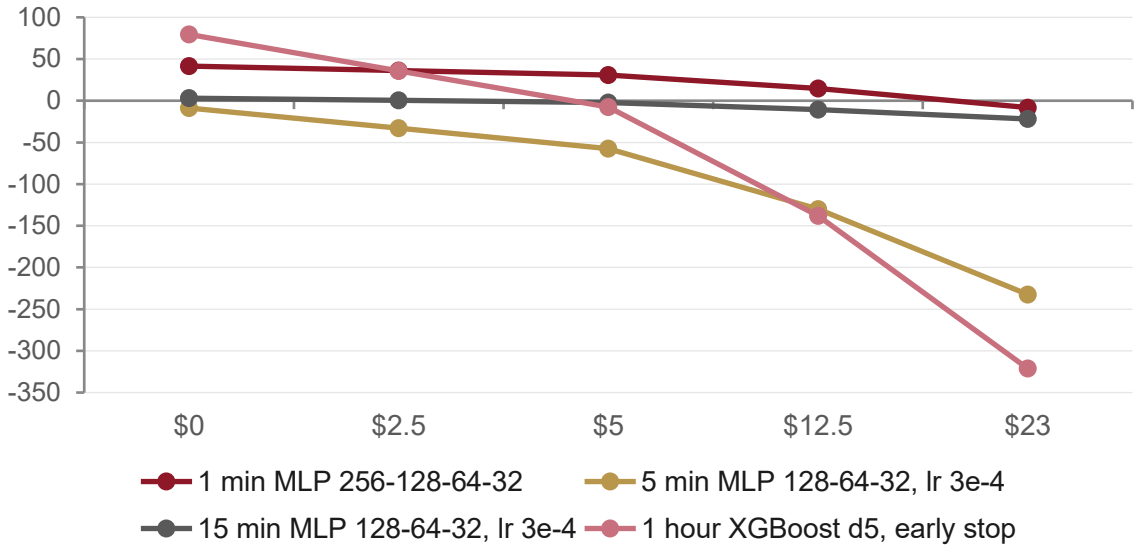
- The model calls “up” 86% of the time, so Long & Short is mostly long and inherits its losses
- Short-only trades far less, so it pays far less in commissions and has the smallest drawdown
- Gross P&L is positive (+$32.6k) but costs (−$107.5k) exceed it; the filter cuts costs, yet the curve ends near zero

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Robustness: Costs, Walk-Forward, In vs Out of Sample

Validation profits shrink or reverse on unseen data, and walk-forward retraining confirms the instability

Test net P&L vs cost per round trip, filtered candidates (thousand USD)



$12.50 and $23 add one tick of spread plus ¼ or 1 tick of slippage per side: realistic market-order execution.

Net P&L by sample and walk-forward years (filtered, $5)

Candle	Train	Validation	Test	Walk-forward years > 0	Walk-forward net
1 min	+$1.7k	+$48.8k	**+$30.7k**	2 of 4	+$3.7k
5 min	−$63.1k	+$81.3k	**−$57.4k**	1 of 4	−$16.7k
15 min	+$6.8k	+$51.0k	**−$2.4k**	1 of 4	−$65.5k
1 hour	+$63.2k	+$133.8k	**−$7.6k**	2 of 4	−$157.5k

Validation picks the best of 349 configurations, so its profit is optimistic by construction (multiple comparisons). Walk-forward: model refitted each year on all prior data, threshold re-chosen on the preceding year.

No candidate has a 95% interval of daily net P&L above zero in the test period, and none is profitable in at least 3 of 4 walk-forward years

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Best Overall Strategy

No deployable NASDAQ trading strategy has been identified

Selected by the pre-specified rule

highest validation profit across all 349 configurations

Model	XGBoost d5, early stop
Candles · direction	1 hour · Long & Short
Confidence filter	|P(up) − 0.5| ≥ 0.01
Test accuracy	51.76%
Net P&L (test)	**−$7.6k**
Trades · win rate	17,419 · 51.7%
Profit factor	1.00
Max drawdown	−$133.2k
Daily net 95% CI	−$242 to $234
Walk-forward	2 of 4 years positive, −$157.5k

Best test-period result (descriptive only)

chosen after seeing the test: not a valid selection

Model	MLP 256-128-64-32
Candles · direction	1 min · Long & Short
Confidence filter	|P(up) − 0.5| ≥ 0.1
Test accuracy	48.71%
Net P&L (test)	**+$30.7k**
Trades · win rate	2,163 · 52.1%
Profit factor	1.10
Max drawdown	−$16.4k
Daily net 95% CI	−$11 to $85
Walk-forward	2 of 4 years positive, +$3.7k

Verdict: neither strategy passes the pre-set bar (test profit at $5 and $12.50, daily-P&L interval above zero, 3 of 4 profitable walk-forward years). The 1 min result also assumes filling at the next open with zero delay. Trading either would be a bet, not an edge.

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Appendix

A1 Data, splits and sample sizes

A2 Features, hyper-parameters and selection rules

A3 Classification metrics and confusion matrix

A4 Net P&L by candle length, model and direction

A5 Why 50% means random

A6 Reproducibility and limitations

A1 Data, Splits and Sample Sizes

Same chronological boundaries for every candle length; test scored once

Candle	Rows (all)	Train	Rows fitted	Validation	Test	Test dojis	Test bullish share
1 min	5,884,299	3,167,361	2,000,000	1,346,246	1,370,692	58,616	50.4%
5 min	1,312,758	770,236	653,403	269,182	273,340	4,933	50.8%
15 min	442,614	264,084	240,005	88,341	90,189	909	51.3%
1 hour	108,026	64,877	62,096	21,642	21,507	104	51.8%

- Train 2006-01-03 → 2017-08-24, validation 2017-08-25 → 2021-07-14, test 2021-07-15 → 2025-05-30 (session dates). Labels crossing a boundary are purged.
- Source: TradeStation continuous NQ, 1-minute bars, Europe/Rome clock, end-stamped; additively back-adjusted, so within-session point moves equal the traded contract's.
- Instrument: CME E-mini Nasdaq-100 futures (NQ), $20 per index point, tick 0.25 = $5. Not MNQ ($2 per point), not QQQ, not the cash index.
- Candles are built from 1-minute bars inside CME sessions; a trade never spans a session close. Bad prints (|1-min return| > 2%) are quarantined causally.
- Training rows are non-doji candles; at 1 minute the most recent 2,000,000 are used. The test period was inspected with the original models in earlier project phases: results are exploratory.

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A2 Features, Hyper-parameters and Selection Rules

All choices made on training and validation data; the test set never informed them

Features (all known at the close of candle t)

Group	Features
Original (8)	log returns lag 1, 2, 3, 5; momentum 5, 10; volatility 5, 10
Candle shape	body / close, high-low range, upper and lower wick
Volume, volatility	log volume vs 20-bar mean; 20-bar realised volatility
Indicators	RSI(14) centred; gap to 20 and 50-bar moving averages
Time	time of day within the session (sine, cosine)

Configurations tested per candle length

Family	Original	Variants (new features)
Logistic	C = 1, 8 features	C = 1 and C = 0.01
Random forest	200 trees, depth 6	200 trees, depth 10, leaf 200
XGBoost	200 trees, depth 3, η 0.05	depth 3 or 5, η 0.05, early stop on validation
MLP	64-32-16, ReLU, L2 1e-3	64-32, 128-64-32, 256-128-64-32; ±dropout 0.2 + batch norm; lr 3e-4 / batch 1024; AdamW, early stop

- Optimised model per family = highest validation balanced accuracy (it may be the original).
- Confidence threshold per model and direction = highest validation net P&L on {0, 0.01, 0.02, 0.03, 0.05, 0.075, 0.10}, at least 200 trades.
- Overall strategy = highest validation net P&L; deployable only if test net > 0 at $5 and $12.50, daily-P&L 95% interval > 0 and ≥ 3 of 4 profitable walk-forward years.
- Scaling fitted on training rows only; network training shuffles only inside the training set; seeds fixed (42).

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A3 Classification Metrics and Confusion Matrix

Test period, 5 min candles: originals and optimised versions

Model	Accuracy	Non-doji	Balanced	Precision (up)	Recall (up)	F1 (up)	Bullish acc.	Bearish acc.	vs majority, pp [95% CI]
Always long	49.90%	50.81%	50.00%	50.81%	100.00%	67.39%	100.00%	0.00%	0.00 [0.00, 0.00]
Reversal rule	50.30%	51.23%	51.22%	52.04%	51.27%	51.65%	51.27%	51.18%	0.40 [0.12, 0.68]
Logistic (original)	50.42%	51.35%	51.16%	51.76%	62.85%	56.76%	62.85%	39.48%	0.53 [0.25, 0.80]
Logistic C=1 + features	50.43%	51.36%	51.28%	51.99%	55.90%	53.87%	55.90%	46.67%	0.53 [0.27, 0.79]
XGBoost (original)	50.50%	51.43%	51.33%	52.00%	57.43%	54.58%	57.43%	45.23%	0.60 [0.32, 0.91]
XGBoost d3, early stop	50.46%	51.39%	51.32%	52.03%	55.73%	53.81%	55.73%	46.91%	0.57 [0.27, 0.86]
Random forest (original)	50.49%	51.42%	51.32%	52.00%	57.33%	54.53%	57.33%	45.32%	0.60 [0.30, 0.88]
Random forest d10	50.47%	51.40%	51.34%	52.07%	54.93%	53.46%	54.93%	47.75%	0.57 [0.28, 0.88]
MLP 64-32-16 (original)	50.33%	51.26%	51.12%	51.78%	59.48%	55.36%	59.48%	42.77%	0.44 [0.16, 0.71]
MLP 128-64-32	50.52%	51.45%	51.42%	52.18%	53.16%	52.67%	53.16%	49.68%	0.62 [0.32, 0.92]

Confusion matrix, MLP 128-64-32 (non-doji candles)

	Predicted up	Predicted down
Actual up	72,499	63,891
Actual down	66,431	65,586

n = 273,340 test candles (136,390 bullish, 132,017 bearish, 4,933 doji). Accuracy ± 0.19 pp (binomial 95%); the paired comparison with the majority rule uses 20-session block bootstrap.

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A4 Net P&L by Candle Length, Model and Direction

Test period, $5 per round trip, thousand USD: unfiltered (left) and with the validation-chosen filter (right)

Candle	Model (optimised)	L&S	Long	Short	L&S filtered	Long filtered	Short filtered
1 min	Logistic C=1 + features	-7715.1	-4023.4	-3691.7	-70.6	-45.5	-25.1
	Random forest d10	-7366.8	-4066.8	-3300.0	-3.3	-3.2	-65.2
	XGBoost d5, early stop	-7393.0	-4122.6	-3270.3	-14.0	-11.5	-2.5
	MLP 256-128-64-32	-7088.5	-3909.7	-3178.8	30.7	2.2	28.5
5 min	Logistic C=1 + features	-1158.5	-630.8	-527.7	36.1	-16.1	44.0
	Random forest d10	-1045.1	-560.1	-485.0	-5.5	10.1	-15.6
	XGBoost d3, early stop	-1135.4	-616.3	-519.1	29.9	14.6	15.3
	MLP 128-64-32	-1138.5	-581.6	-556.9	-26.1	-3.5	-9.8
15 min	Logistic C=0.01 + features	-557.1	-317.7	-239.4	-16.6	-36.6	20.0
	Random forest d10	-335.6	-200.6	-135.0	4.0	3.8	13.7
	XGBoost d3, early stop	-408.7	-251.7	-157.0	-15.6	-18.1	44.8
	MLP 128-64-32	-515.1	-290.3	-224.9	-52.5	-60.8	8.2
1 hour	Logistic C=1 + features	-128.9	-89.2	-39.8	-189.4	-201.3	-20.0
	Random forest d10	-5.2	-31.0	25.7	-61.4	—	-0.5
	XGBoost d5, early stop	-74.9	-69.1	-5.8	-7.6	-29.1	21.5
	MLP 128-64-32	-130.0	-92.5	-37.6	-137.6	-106.9	-30.7

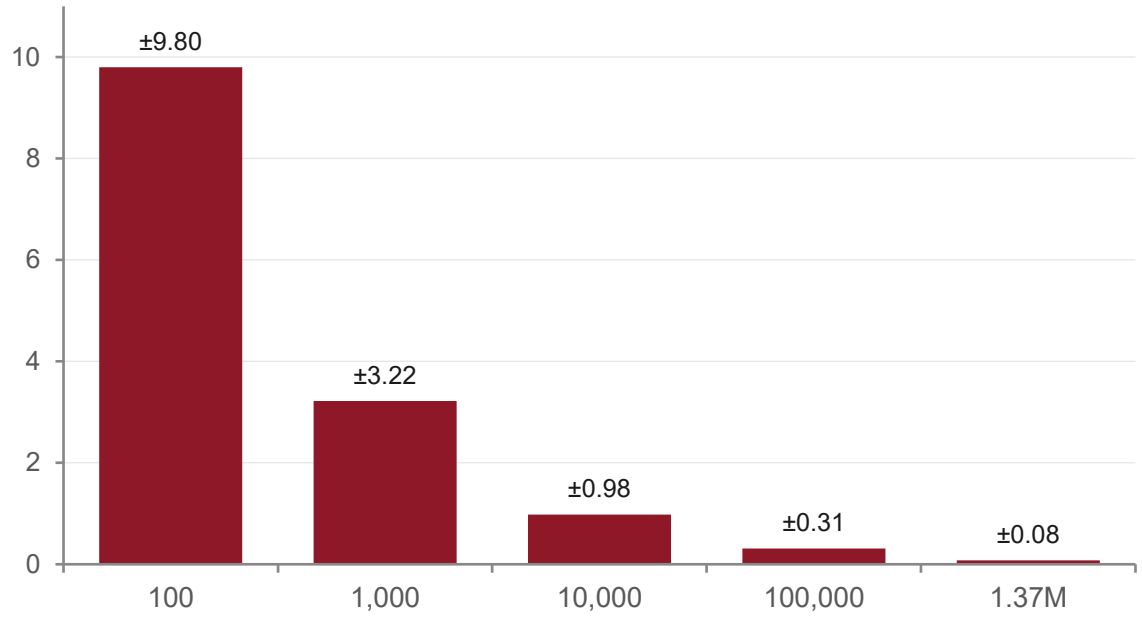
"—": the validation-chosen threshold was 0 (no filter) or produced fewer than 200 validation trades. Filtered results are spread around zero with no consistent sign across candle lengths or models.

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A5 Why 50% Means Random

Predicting the next candle is close to tossing a coin; only results outside luck's range count

How far luck alone moves a fair coin's accuracy (95% range, ± pp)



Range = 1.96 × √(0.25 / n). With few trades a coin can look like a 55–60% "model"; with a million it cannot leave 50 ± 0.1%.

Reading accuracy around 50%

1 **Coin toss = 50%**
if up and down are equally likely and unpredictable, any rule is right half the time

2 **Dojis lower the bar**
candles with close = open count as misses, so all-candle accuracy can sit below 50%

3 **Imbalance is not skill**
NQ rose over time, so "always up" already beats a coin without predicting anything

4 **Accuracy ≠ money**
a 51% hit rate earns nothing if wins and losses are equal in size and each trade costs $5

Best model: 51.45% on non-doji 5 min candles: statistically real, economically too small

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A6 Reproducibility and Limitations

Every number in this deck is regenerated and cross-checked by four commands from the public repository

Reproduce (Windows PowerShell, project folder)

Repository: github.com/Paciooo/Project_ai_ML_bigdata

```
$env:FUTURES_DATA_DIR="C:\path\to\futures-data"
```

1. `python study.py`

   trains all models, backtests, walk-forward (about 15 min on 16 CPU cores, no GPU)

2. `python report.py results/nq_study/runs/<run> presentation/deck_data.json`

3. `python verify.py results/nq_study/runs/<run>`

4. `cd presentation; npm install; node build_deck.js`

Run used here: 20261111T070520Z_949daa_final (commit ff445f3), seed 42, latency 0s, $5 per round trip.

Limitations

- Idealised execution: fill at the next open with zero delay and zero slippage, 100% limit fills; a live limit order may not fill.
- Bar prices, not bid/ask quotes; costs are assumptions. $12.50 / $23 scenarios approximate market orders.
- Continuous back-adjusted series; roll dates are approximate (no position spans a session).
- The 2021–2025 test period was seen in earlier phases; data after May 2025 (reserved) was not available.
- Daily Sharpe ratios use 252 sessions; percentages use an assumed $100,000 account.

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